



Qualification Pack



Continuous Casting - Process Control Engineer

QP Code: ISC/Q1405

Version: 1.0

NSQF Level: 6

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ISC/Q1405: Continuous Casting - Process Control Engineer

Brief Job Description

The individual in this role coordinate and monitor the continuous casting operation and its associated functions.

Personal Attributes

The person should be patient, organized, team-oriented and have the ability to work for long hours in adverse conditions. They must be able to plan and prioritise tasks effectively and have an eye for detail and quality.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ISC/N0946: Manage and implement health and safety practices at the work place](#)
2. [ISC/N0947: Manage team and others at workplace](#)
3. [ISC/N1405: Prepare for Continuous Casting Operation](#)
4. [ISC/N1411: Perform and control continuous casting operations](#)
5. [ISC/N1704: Apply Decarbonization practice in plant](#)
6. [ISC/N1703: Apply statistical process control \(SPC\) techniques](#)
7. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Steel Making
Country	India
NSQF Level	6
Credits	20



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Aligned to NCO/ISCO/ISIC Code	NCO-2015/3122.4401
Minimum Educational Qualification & Experience	Completed 4 year UG program (BE/B. Tech in Metallurgy / Material Science / Mechanical / Production Engineering) OR Completed 2nd year diploma after 12th (in Metallurgy / Material Science / Mechanical / Production Engineering) with 3 Years of experience OR Completed 3 year diploma after 10th (in Metallurgy / Material Science / Mechanical / Production Engineering) with 3-5 Years of experience 4.5 Years OR Previous relevant Qualification of NSQF Level (5) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	30/04/2027
NSQC Approval Date	30/04/2024
Version	1.0
Reference code on NQR	QG-06-IS-02477-2024-V1-IISSSC
NQR Version	1



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ISC/N0946: Manage and implement health and safety practices at the work place

Description

This OS unit is about managing and implementing health, hygiene, safety and sustainable practices for optimising use of resources.

Scope

The scope covers the following :

- Maintain safe and secure working environment
- Maintain health and hygiene
- Manage housekeeping at workplace
- Implement material and energy conservation practices

Elements and Performance Criteria

Maintain safe and secure working environment

To be competent, the user/individual on the job must be able to:

- PC1.** identify hazardous activities and the possible causes of risks or accidents in the workplace
- PC2.** implement safe working practices while dealing with hazards to ensure safety of self and others
- PC3.** ensure that team is following appropriate safety practices while working in and around trenches, elevated places and confined areas
- PC4.** conduct regular checks of the machines with support of the maintenance team to identify potential hazards
- PC5.** organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices
- PC6.** fill daily check sheet to report improvements done and risks identified
- PC7.** ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others
- PC8.** report any identified breaches in health, safety and security policies and procedures to the designated person
- PC9.** ensure that team is following appropriate emergency procedures such as raising alarm, safe evacuation etc.

Maintain health and hygiene

To be competent, the user/individual on the job must be able to:

- PC10.** ensure team is aware about hygiene and sanitation regulations and following them on the shop floor
- PC11.** ensure workplace, equipment, restrooms etc. are cleaned and sanitized regularly
- PC12.** ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace
- PC13.** report advanced hygiene and sanitation issues to appropriate authority



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PC14. report regarding the contagious illness of self or people in close contact

Manage housekeeping at workplace

To be competent, the user/individual on the job must be able to:

PC15. implement appropriate housekeeping methods and practices at workplace

PC16. ensure good housekeeping in order to prevent hazards and accidents

PC17. monitor that that team is storing the material, tools and equipment in the correct location and in good condition after work completion

PC18. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP

PC19. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste

Implement material and energy conservation practices

To be competent, the user/individual on the job must be able to:

PC20. identify processes where material and energy/electricity utilization can be optimized

PC21. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively

PC22. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water

PC23. identify possibilities of using renewable energy and environment friendly fuels

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organisation procedures for health, safety and security, individual role and responsibilities in this context

KU2. the organisation's emergency procedures for different emergency situations and the importance of following the same

KU3. evacuation procedures for workers and visitors

KU4. how and when to report hazards

KU5. potential hazards, risks and threats based on the nature of work

KU6. preventative and remedial actions to be taken in case of exposure to toxic material

KU7. various types of fire extinguisher

KU8. various types of safety signs and their meaning

KU9. appropriate first aid treatment relevant to different condition e.g. bleeding, minor burns, eye injuries etc.

KU10. relevant standards, procedures and policies related to 5S followed in the company

KU11. the various materials used and their storage norms

KU12. efficient utilisation of material and water

KU13. basics of electricity and prevalent energy efficient devices

KU14. common practices of conserving electricity

KU15. categorisation of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics

KU16. usage of different colors of dustbins



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KU17. waste management techniques

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read safety instructions/guidelines
- GS2.** modify work practices to improve them
- GS3.** work with supervisors/team members to carry out work related tasks
- GS4.** complete tasks efficiently and accurately within stipulated time
- GS5.** inform/report to concerned person in case of any problem
- GS6.** make timely decisions for efficient utilization of resources
- GS7.** write reports such as accident report, in at least English/regional language



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain safe and secure working environment</i>	12	21	-	8
PC1. identify hazardous activities and the possible causes of risks or accidents in the workplace	2	3	-	1
PC2. implement safe working practices while dealing with hazards to ensure safety of self and others	2	2	-	1
PC3. ensure that team is following appropriate safety practices while working in and around trenches, elevated places and confined areas	1	2	-	-
PC4. conduct regular checks of the machines with support of the maintenance team to identify potential hazards	2	4	-	1
PC5. organise safety drills or training sessions to create awareness amongst others on the identified risks and safety practices	1	2	-	1
PC6. fill daily check sheet to report improvements done and risks identified	1	2	-	1
PC7. ensure that relevant safety boards/signs are placed on the shop floor for the safety of self and others	1	2	-	1
PC8. report any identified breaches in health, safety and security policies and procedures to the designated person	1	2	-	1
PC9. ensure that team is following appropriate emergency procedures such as raising alarm, safe evacuation etc.	1	2	-	1
<i>Maintain health and hygiene</i>	6	9	-	4
PC10. ensure team is aware about hygiene and sanitation regulations and following them on the shop floor	2	3	-	1
PC11. ensure workplace, equipment, restrooms etc. are cleaned and sanitized regularly	1	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure availability of running water, hand wash and alcohol-based sanitizers at the workplace	1	2	-	1
PC13. report advanced hygiene and sanitation issues to appropriate authority	1	1	-	1
PC14. report regarding the contagious illness of self or people in close contact	1	1	-	-
<i>Manage housekeeping at workplace</i>	6	10	-	4
PC15. implement appropriate housekeeping methods and practices at workplace	2	3	-	1
PC16. ensure good housekeeping in order to prevent hazards and accidents	1	2	-	1
PC17. monitor that that team is storing the material, tools and equipment in the correct location and in good condition after work completion	1	2	-	1
PC18. ensure recyclable, non-recyclable and hazardous wastes are segregated as per SOP	1	2	-	1
PC19. ensure proper mechanism is followed while collecting and disposing of non-recyclable, recyclable and reusable waste	1	1	-	-
<i>Implement material and energy conservation practices</i>	6	10	-	4
PC20. identify processes where material and energy/electricity utilization can be optimized	2	3	-	1
PC21. ensure malfunctioning (fumes/sparks/emission/vibration/noise) and lapse in maintenance of equipment are resolved effectively	1	2	-	1
PC22. prepare and analyze material and energy audit reports to decipher excessive consumption of material and water	2	3	-	1
PC23. identify possibilities of using renewable energy and environment friendly fuels	1	2	-	1
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0946
NOS Name	Manage and implement health and safety practices at the work place
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance, Raw Material Handling , Product Quality Control, Refractory Maintenance, Furnace Operation, Fitter, Cold Rolling Mill Operations, Electrical Maintenance
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



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ISC/N0947: Manage team and others at workplace

Description

This OS unit is about managing team, colleagues/superiors and others, either in own work group or in other work groups within organisation.

Scope

The scope covers the following :

- Communicate effectively with team members
- Interact with supervisor
- Manage appropriate behaviour at work place

Elements and Performance Criteria

Communicate effectively with team members

To be competent, the user/individual on the job must be able to:

- PC1.** manage and co-ordinate with team members to integrate work, as per the workload in order to achieve team goals
- PC2.** implement ways to share information with team members in line with organisational requirements
- PC3.** ensure that work requirements are clearly communicated to the team members through all means including face-to-face, telephonic and written
- PC4.** work in a way that show respect for all team members and customers
- PC5.** carry out commitments made to team members
- PC6.** resolve conflicts within the team members at work to achieve smooth workflow
- PC7.** guide the team members to follow the organisation's policies and procedures
- PC8.** ensure team goals are given preference over individual goals
- PC9.** encourage team members to rectify errors as per feedback and minimize mistakes in future

Interact with supervisor

To be competent, the user/individual on the job must be able to:

- PC10.** report progress on job allocated and team performance to the superiors
- PC11.** escalate problems to supervisors that cannot be handled
- PC12.** report the completed work
- PC13.** train the team members to report completed work and receive feedback on work done

Manage appropriate behaviour at work place

To be competent, the user/individual on the job must be able to:

- PC14.** help PwD team members to overcome the challenges, if asked
- PC15.** ensure team shows sensitivity towards all genders and PwD
- PC16.** adjust communication styles to reflect gender sensitivity and sensitivity towards person with disability



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PC17. ensure that team is following responsible and disciplined behaviours at the workplace

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of effective communication and establishing good working relationships with colleagues and supervisor
- KU2.** different methods of communication as per the circumstances
- KU3.** importance of teamwork in organization and individual success
- KU4.** various components of effective communication
- KU5.** barriers to effective communication
- KU6.** common reasons for interpersonal conflict
- KU7.** what constitutes disciplined behaviour for a working professional
- KU8.** gender concepts, issues & legislation
- KU9.** organisational policies and procedures related to gender equality
- KU10.** challenges faced by PWD and the ways to help them overcome the same

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions/guidelines/procedures
- GS2.** listen effectively and orally communicate information
- GS3.** ask for clarification and advice from the concerned person
- GS4.** maintain positive and effective relationships with colleagues
- GS5.** evaluate the possible solution(s) to the problem
- GS6.** spot and communicate potential areas of disruptions in the work process and report the same
- GS7.** complete written work with attention to detail
- GS8.** check that the work meets customer requirements



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with team members</i>	14	24	-	10
PC1. manage and co-ordinate with team members to integrate work, as per the workload in order to achieve team goals	3	5	-	2
PC2. implement ways to share information with team members in line with organisational requirements	3	5	-	2
PC3. ensure that work requirements are clearly communicated to the team members through all means including face-to-face, telephonic and written	2	4	-	2
PC4. work in a way that show respect for all team members and customers	1	2	-	1
PC5. carry out commitments made to team members	1	1	-	1
PC6. resolve conflicts within the team members at work to achieve smooth workflow	1	2	-	-
PC7. guide the team members to follow the organisation's policies and procedures	1	2	-	1
PC8. ensure team goals are given preference over individual goals	1	1	-	1
PC9. encourage team members to rectify errors as per feedback and minimize mistakes in future	1	2	-	-
<i>Interact with supervisor</i>	8	10	-	6
PC10. report progress on job allocated and team performance to the superiors	2	2	-	1
PC11. escalate problems to supervisors that cannot be handled	2	3	-	2
PC12. report the completed work	2	2	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. train the team members to report completed work and receive feedback on work done	2	3	-	2
<i>Manage appropriate behaviour at work place</i>	8	16	-	4
PC14. help PwD team members to overcome the challenges, if asked	2	4	-	1
PC15. ensure team shows sensitivity towards all genders and PwD	2	4	-	1
PC16. adjust communication styles to reflect gender sensitivity and sensitivity towards person with disability	2	4	-	1
PC17. ensure that team is following responsible and disciplined behaviours at the workplace	2	4	-	1
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0947
NOS Name	Manage team and others at workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory, Mechanical Maintenance, Iron Making
Occupation	Mechanical Maintenance, Raw Material Handling , Coke Making, Operations, Iron Making, Cold Rolling Mill Operations, Product Quality Control, Electrical Maintenance, Electronics and Instrumentation Maintenance, Refractory Maintenance, Furnace Operation, Fitter, Steel Making, Pre-fabricated Structure
NSQF Level	5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N1405: Prepare for Continuous Casting Operation

Description

This OS unit is about performing preparatory activities for carrying out various continuous casting operations

Scope

The scope covers the following :

- Carry out preparatory activities for continuous casting operation

Elements and Performance Criteria

Carry out preparatory activities for continuous casting operation

To be competent, the user/individual on the job must be able to:

- PC1.** obtain the required information and work completed in previous shift from logbook, discussions etc. and take handover from the previous shift operator by following organizational procedures
- PC2.** ensure availability of sufficient quantity of raw material required for smooth operations and achieving production target
- PC3.** ensure that team is checking the quality of raw materials
- PC4.** ensure that work area is clean and free of any obstacle or encumbrance
- PC5.** ensure that tundish is heating at the required temperature
- PC6.** carry out machine setting with mould and water-cooling system
- PC7.** ensure that all safety measures, interlocks, alarms are working and in conformance with the safe operation of casting plant as per SOP
- PC8.** coordinate with other related departments such as raw material handling and quality departments

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** different types of casting processes
- KU3.** Standard Operating Procedures (SOP) recommended by OEM for using tools and equipment required during casting operations
- KU4.** basic process of ladle furnace and continuous casting process
- KU5.** material required for continuous casting process
- KU6.** SOP as recommended by competent authority of the organisation
- KU7.** normal running characteristics of continuous casting equipment
- KU8.** impact of operating parameters i.e casting speed, material flow, cooling water flow and pressure etc.



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- KU9.** effect of quality of raw material on the process
- KU10.** emergency procedures and safety measures such as gas safety, hot metal handling safety etc. associated with continuous casting operation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out preparatory activities for continuous casting operation</i>	30	50	-	20
PC1. obtain the required information and work completed in previous shift from logbook, discussions etc. and take handover from the previous shift operator by following organizational procedures	3	5	-	3
PC2. ensure availability of sufficient quantity of raw material required for smooth operations and achieving production target	3	5	-	2
PC3. ensure that team is checking the quality of raw materials	3	5	-	2
PC4. ensure that work area is clean and free of any obstacle or encumbrance	4	6	-	2
PC5. ensure that tundish is heating at the required temperature	4	7	-	3
PC6. carry out machine setting with mould and water-cooling system	5	9	-	3
PC7. ensure that all safety measures, interlocks, alarms are working and in conformance with the safe operation of casting plant as per SOP	4	7	-	3
PC8. coordinate with other related departments such as raw material handling and quality departments	4	6	-	2
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1405
NOS Name	Prepare for Continuous Casting Operation
Sector	Iron and Steel
Sub-Sector	
Occupation	Steel Making, Steel Making
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



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ISC/N1411: Perform and control continuous casting operations

Description

This OS unit is about performing various continuous casting operations as per the final output specifications and the standards specified by the organization

Scope

The scope covers the following :

- Perform continuous casting operations
- Perform post-operation activities

Elements and Performance Criteria

Perform continuous casting process

To be competent, the user/individual on the job must be able to:

- PC1.** use appropriate Personal Protective Equipment (PPE) for safe working in work area
- PC2.** check the temperature of liquid steel in the ladle and ensure that it is as per the requirement
- PC3.** place the ladle on the continuous casting platform
- PC4.** open the ladle nozzle and fill up the tundish at required level
- PC5.** open the tundish nozzle and fill the liquid steel in the mould upto desired level with mould water cooling
- PC6.** adjust the casting speed, oscillation, frequency, negative strip control etc. as per the section cast during the casting
- PC7.** maintain the flow of liquid steel as per the casting speed
- PC8.** maintain the flow, pressure and back pressure of mould cooling water as per the requirement
- PC9.** adjust the flow of cooling water in secondary zone
- PC10.** control feeding of casting powder in the mould during casting process
- PC11.** monitor that team is cutting the billets in required shape and size by following SOP and appropriate method
- PC12.** send the sample for quality check i.e. chemical composition, rhombicity, surface cracks, macros etc. by following organizational procedure

Perform post-operation activities

To be competent, the user/individual on the job must be able to:

- PC13.** update and maintain shift log of production and prepare production and other reports
- PC14.** record all instructions received from higher authorities in the instruction book and all incidents in the occurrence book
- PC15.** inform concerned authority in case of any abnormalities observed during execution of work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** relevant standards and procedures followed in the company
- KU2.** different types of casting processes
- KU3.** Standard Operating Procedures (SOP) recommended by OEM for using tools and equipment required during casting operations
- KU4.** basic process of ladle furnace and continuous casting process
- KU5.** material required for continuous casting process
- KU6.** SOP as recommended by competent authority of the organisation
- KU7.** normal running characteristics of continuous casting equipment
- KU8.** impact of operating parameters i.e casting speed, material flow, cooling water flow and pressure etc.
- KU9.** effect of quality of raw material on the process
- KU10.** procedure of operating continuous casting machine and equipment
- KU11.** how to monitor flow of liquid steel
- KU12.** how the quality parameters of liquid steel affect downstream process productivity, quality and cost point of view
- KU13.** post continuous casting operation activities i.e. inspection, reporting etc.
- KU14.** quality check process of casted product
- KU15.** emergency procedures and safety measures such as gas safety, hot metal handling safety etc. associated with continuous casting operation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform continuous casting process</i>	25	42	-	17
PC1. use appropriate Personal Protective Equipment (PPE) for safe working in work area	-	1	-	-
PC2. check the temperature of liquid steel in the ladle and ensure that it is as per the requirement	3	4	-	1
PC3. place the ladle on the continuous casting platform	2	4	-	1
PC4. open the ladle nozzle and fill up the tundish at required level	2	4	-	1
PC5. open the tundish nozzle and fill the liquid steel in the mould upto desired level with mould water cooling	2	4	-	1
PC6. adjust the casting speed, oscillation, frequency, negative strip control etc. as per the section cast during the casting	3	5	-	3
PC7. maintain the flow of liquid steel as per the casting speed	2	3	-	2
PC8. maintain the flow, pressure and back pressure of mould cooling water as per the requirement	2	3	-	2
PC9. adjust the flow of cooling water in secondary zone	2	3	-	1
PC10. control feeding of casting powder in the mould during casting process	2	3	-	1
PC11. monitor that team is cutting the billets in required shape and size by following SOP and appropriate method	3	5	-	2
PC12. send the sample for quality check i.e. chemical composition, rhombicity, surface cracks, macros etc. by following organizational procedure	2	3	-	2
<i>Perform post-operation activities</i>	5	8	-	3



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. update and maintain shift log of production and prepare production and other reports	2	3	-	2
PC14. record all instructions received from higher authorities in the instruction book and all incidents in the occurrence book	2	3	-	1
PC15. inform concerned authority in case of any abnormalities observed during execution of work	1	2	-	-
NOS Total	30	50	-	20



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1411
NOS Name	Perform and control continuous casting operations
Sector	Iron and Steel
Sub-Sector	
Occupation	Steel Making
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024



Qualification Pack

ISC/N1704: Apply Decarbonization practice in plant

Description

This OS unit is about apply decarbonization practices in steel plant to reduce carbon emissions.

Scope

The scope covers the following :

- Apply decarbonization practices

Elements and Performance Criteria

Apply decarbonization practices

To be competent, the user/individual on the job must be able to:

- PC1.** identify various processes (i.e. BOF, DRI, EAF etc.) and fuels (i.e. oil, coal etc.) used in plant during production
- PC2.** calculate carbon emission by summing the emissions resulting from every stage of a product or service's lifetime (material production, manufacturing, use, and end-of-life)
- PC3.** implement efficient technologies and processes in production process that reduce energy use and emissions significantly to achieve net-zero carbon emission in plant
- PC4.** implement technologies such as Carbon capture, usage and storage (CCUS) in production process to reduce and capture carbon emissions in the steelmaking process
- PC5.** replace fossil fuels in BOF, DRI etc. with low/no carbon options such as biofuels, natural gas, hydrogen etc.
- PC6.** use energy sources such as wind, solar, nuclear, and water power combined with advances in electricity storage for minimal CO₂ emissions
- PC7.** use equipment that are powered by electricity instead of fossil fuels
- PC8.** reuse or recycle scrap and waste generated during steelmaking process
- PC9.** switch to renewable electricity and electrifying their vehicle fleets
- PC10.** increase the usage of recycled materials
- PC11.** avoid single use plastics and paper products

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant standards and procedures followed in the company
- KU2.** various fuels such as oil, coal etc. responsible for carbon emission in steel plant
- KU3.** various processes such as BOF, DRI, EAF, rotary kiln etc. responsible for carbon emission in steel plant
- KU4.** gases generated in carbon emission i.e. carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) etc.
- KU5.** CO₂ emission and its effect on environment



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- KU6.** Five R's refuse, reduce, reuse, rot, recycle
- KU7.** Carbon capture, usage and storage (CCUS) technology
- KU8.** carbon dioxide removal (CDR) or negative emissions techniques (NETs)
- KU9.** how to calculate CO2 using different fuels
- KU10.** how to control burning of fuel
- KU11.** renewable energy sources such as wind, solar, nuclear, and water

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions, equipment manuals and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Apply decarbonization practices</i>	30	50	-	20
PC1. identify various processes (i.e. BOF, DRI, EAF etc.) and fuels (i.e. oil, coal etc.) used in plant during production	3	5	-	2
PC2. calculate carbon emission by summing the emissions resulting from every stage of a product or service's lifetime (material production, manufacturing, use, and end-of-life)	3	5	-	2
PC3. implement efficient technologies and processes in production process that reduce energy use and emissions significantly to achieve net-zero carbon emission in plant	3	5	-	2
PC4. implement technologies such as Carbon capture, usage and storage (CCUS) in production process to reduce and capture carbon emissions in the steelmaking process	3	5	-	2
PC5. replace fossil fuels in BOF, DRI etc. with low/no carbon options such as biofuels, natural gas, hydrogen etc.	3	5	-	2
PC6. use energy sources such as wind, solar, nuclear, and water power combined with advances in electricity storage for minimal CO2 emissions	3	5	-	2
PC7. use equipment that are powered by electricity instead of fossil fuels	3	5	-	2
PC8. reuse or recycle scrap and waste generated during steelmaking process	3	5	-	2
PC9. switch to renewable electricity and electrifying their vehicle fleets	2	4	-	2
PC10. increase the usage of recycled materials	2	3	-	1
PC11. avoid single use plastics and paper products	2	3	-	1
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1704
NOS Name	Apply Decarbonization practice in plant
Sector	Iron and Steel
Sub-Sector	
Occupation	Generic
NSQF Level	6
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N1703: Apply statistical process control (SPC) techniques

Description

This OS unit is about apply statistical process control (SPC) technique to improve the product quality and production efficiency.

Scope

The scope covers the following :

- Apply statistical process control (SPC) technique

Elements and Performance Criteria

Apply statistical process control (SPC) techniques

To be competent, the user/individual on the job must be able to:

- PC1.** identify the processes that impacts the output of the product
- PC2.** identify the attributes that need to be measured during production
- PC3.** develop a subgroup strategy and sampling plan based on the product's criticality
- PC4.** collect the data per sample size and select an appropriate SPC chart based on data type (Continuous or Discrete) and subgroup size
- PC5.** calculate standard deviation and variance of the production
- PC6.** prepare normal distribution curve on the basis of data collected
- PC7.** calculate the upper control limit (UCL) and lower control limit (LCL) to describe the natural variation of attributes
- PC8.** calculate process capabilities by applying standard formula
- PC9.** calculate process capability index and draw graph on the basis of it
- PC10.** identify improvements required in process on the basis of observations of process capability index results
- PC11.** monitor process variation by interpreting the control chart and check whether any point is out of control and the pattern

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** six sigma concept
- KU2.** statistical process control (SPC) technique and its importance
- KU3.** SPC concept i.e. mode, mean, median, mad
- KU4.** benefits of SPC
- KU5.** objectives of SPC
- KU6.** various tools of SPC
- KU7.** how to perform SPC



Qualification Pack

- KU8.** standard deviation and variance
- KU9.** how to prepare normal distribution curve
- KU10.** Process capability index and its calculation formula
- KU11.** how to prepare control charts
- KU12.** concept of accuracy and precision

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read work instructions and process documents
- GS2.** communicate the process requirements to the supervisor and co-workers
- GS3.** attentively listen and comprehend the information given by the supervisor/team members
- GS4.** write work related information in English/regional language
- GS5.** recognize a workplace problem and take suitable action
- GS6.** plan and organize work according to the work requirements
- GS7.** complete the assigned tasks with minimum supervision
- GS8.** report to the supervisor or deal with a colleague individually, depending on the type of concern



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Apply statistical process control (SPC) techniques</i>	30	50	-	20
PC1. identify the processes that impacts the output of the product	2	3	-	1
PC2. identify the attributes that need to be measured during production	2	3	-	1
PC3. develop a subgroup strategy and sampling plan based on the product's criticality	3	5	-	2
PC4. collect the data per sample size and select an appropriate SPC chart based on data type (Continuous or Discrete) and subgroup size	3	5	-	2
PC5. calculate standard deviation and variance of the production	3	5	-	2
PC6. prepare normal distribution curve on the basis of data collected	3	5	-	2
PC7. calculate the upper control limit (UCL) and lower control limit (LCL) to describe the natural variation of attributes	3	5	-	2
PC8. calculate process capabilities by applying standard formula	3	5	-	2
PC9. calculate process capability index and draw graph on the basis of it	3	5	-	2
PC10. identify improvements required in process on the basis of observations of process capability index results	2	4	-	2
PC11. monitor process variation by interpreting the control chart and check whether any point is out of control and the pattern	3	5	-	2
NOS Total	30	50	-	20



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N1703
NOS Name	Apply statistical process control (SPC) techniques
Sector	Iron and Steel
Sub-Sector	
Occupation	Generic
NSQF Level	6
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment



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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms



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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services



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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.
6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70



Qualification Pack

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0946.Manage and implement health and safety practices at the work place	30	50	-	20	100	10
ISC/N0947.Manage team and others at workplace	30	50	-	20	100	10
ISC/N1405.Prepare for Continuous Casting Operation	30	50	-	20	100	20
ISC/N1411.Perform and control continuous casting operations	30	50	-	20	100	30
ISC/N1704.Apply Decarbonization practice in plant	30	50	0	20	100	10
ISC/N1703.Apply statistical process control (SPC) techniques	30	50	0	20	100	10
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
Total	200	330	0	120	650	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.